



LIVE PRESENTATION - MARKETING CAMPAIGN CUSTOMER SEGMENTATION

KINDONG LAWSAMBA
GREAT LERNING

AGENDA



Executive Summary

- What are the key takeaways?
- What are the key next steps?

Problem and Solution Summary

- What problem was being solved?
- Final proposed solution design
- Why is this a 'valid' solution that is likely to solve the problem?

Recommendations for implementation

- What are some key recommendations to implement the solution?
- What are the key actionable for stakeholders?
- What is the expected benefit and/or costs?
- What are the key risks and challenges?

EXECUTIVE SUMMARY



Objective:

The aim is to come up with the best possible customer segments using This is done using the Unsupervised Learning ideas such as Dimensionality Reduction and Clustering.

Data Summary: The dataset is provided with customer demographic details, purchase behaviours, response to previous purchases, complaints, customer offer acceptant, amount spend on products, number of customers provided, marketing strategies.

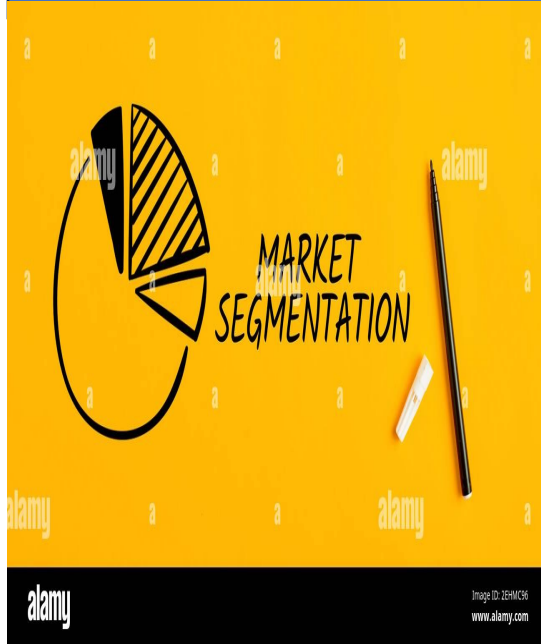
- The dataset consists of 2240 rows and 27 columns.
- Out of 27 columns, 24 columns are numeric, and 3 columns are of object data type.
- There are a few missing values (24) in the Income variable. And no duplicate variables.
- The ID column is an identifier which is unique for each customer.

Best Solutions:

Here are some of the most important reasons why your business needs to segment customers based on their characteristics:

- **More Personalized Marketing Efforts:** Personalized marketing goes beyond just addressing the customer by name. With customer segmentation tools, you can create content and offers based on specific interests and behaviors.
- **Maximized Customer Engagement:** If a segment shows a preference for eco-friendly products, then campaigns can be designed to emphasize sustainability.
- **Lower Advert Expenses:** Online ads are always a huge part of marketing campaigns, but they may not be effective if you don't have a good understanding of who your customers are.
- **Personalized Communication Channels:** Different customers have different preferences for how they receive marketing communications. Some may respond better to emails, while others prefer the immediacy of text messages or the visual appeal of social media.

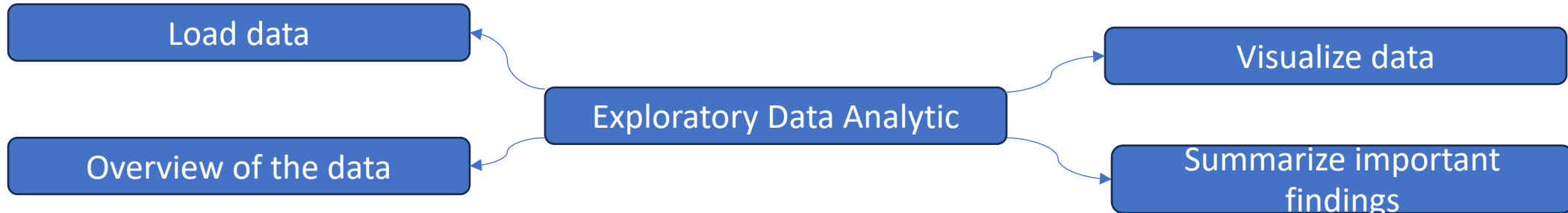
PROBLEM AND SOLUTIONS SUMMARY



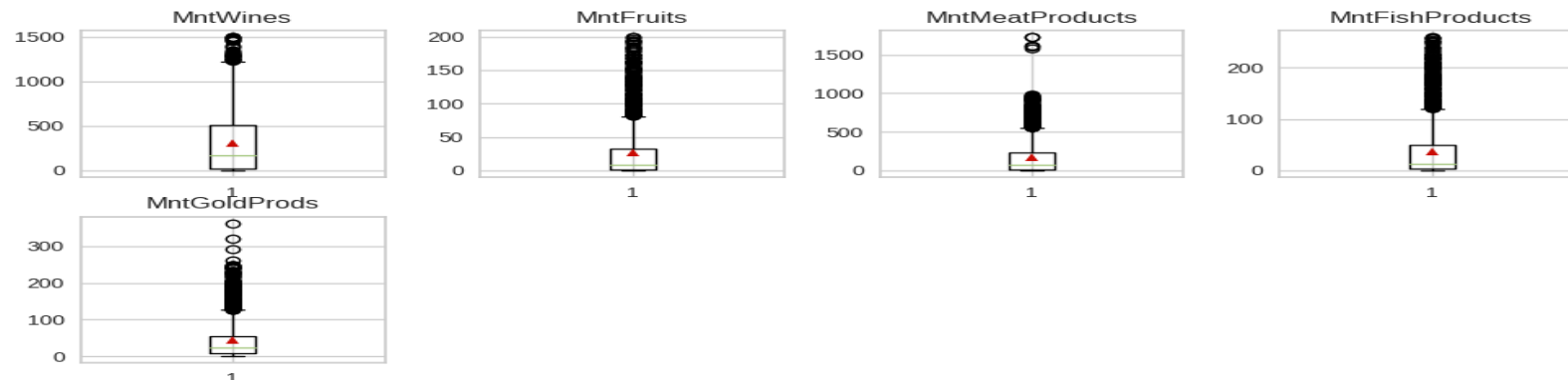
PROBLEM DEFINATION AND OBJECTIVES:

- Customer segmentation is the process of dividing a dataset of customers into groups of similar customers based on certain common characteristics, usually for the purpose of understanding the population dataset in a better fashion.
- Understanding customer behavior and characteristics is usually a critical part of the marketing operations of any business or organization, with direct consequences on sales & marketing strategy.
- Customer segmentation is often viewed to achieve a better return on investment from marketing efforts and make organizations more efficient in terms of utilizing their money, time, and other critical resources in custom marketing strategies for different groups of customers based on their unique needs and motivations.
- Using Unsupervised Learning ideas such as Dimensionality Reduction and Clustering, is to come up with the best possible customer segments using the given customer dataset.

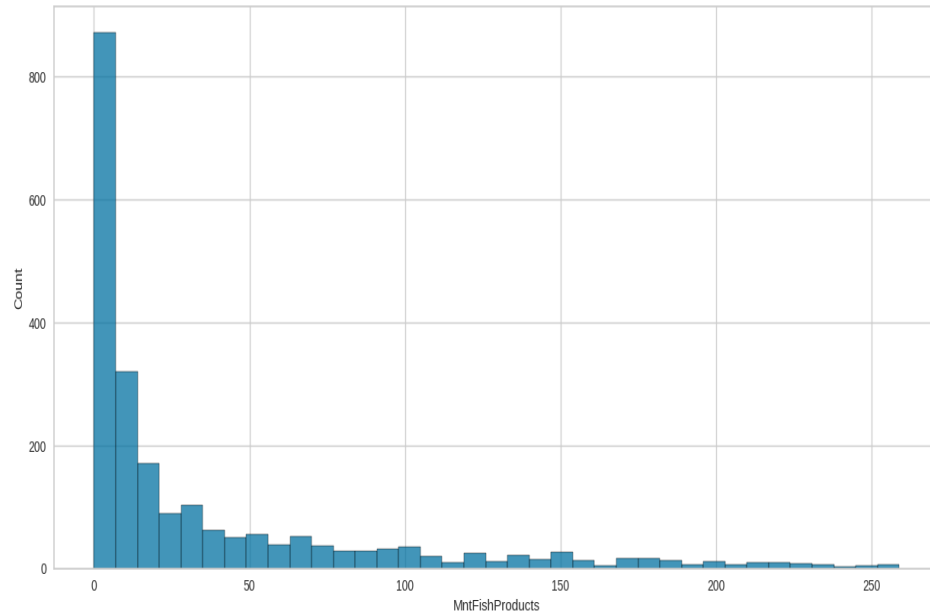
Solution Step 1:- Exploratory Data Analytic(EDA)



- There are 24 missing values for the Income variable with a percentage of 1% in the dataset. There in this scenario dropping the row of missing values will be a good idea as not much data is lost.
- The shape of the dataset remains the same after treating the missing values. That is 2240 rows and 27 columns (24 numerical and 3 object datatype).
- According to the visualizing diagram we can see that there are a lot of outliers indicating extreme values in the features.



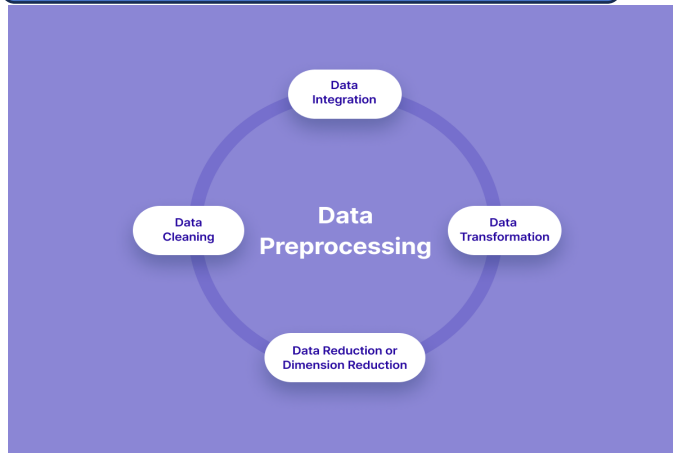
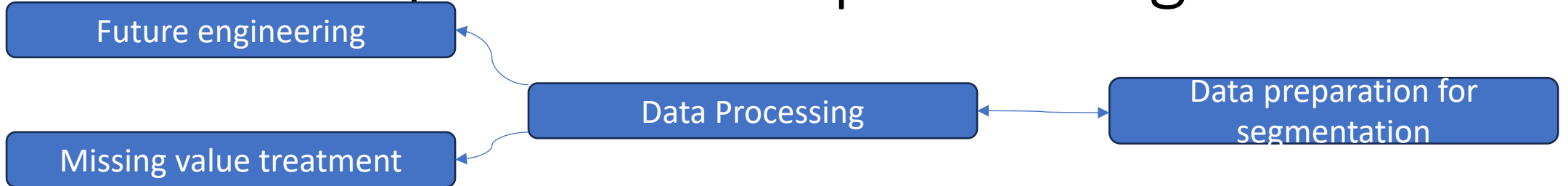
Marketing Campaign Customer Segmentation



- We observed a skew distribution from the features in the dataset, as show in the MntFishProducts diagram.
- We observe that all the numeric features in the dataset have outliers present in them, except income which is not clearly visible.
- We can see that the whiskers are fairly close in the box of MinGoldProds that all the other features. And several of the values above it are clearly outliers.

- The ID column is an identifier which is unique for each customer. That is identifiers are short codes that permanently in dataset. We observe that ID has null values.
- The number of unique values are equal to the number of observations. So the ID column looks like an index for the data entry and a column would not be useful in providing any predictive power for our analysis. Hence, it can be dropped.

Solution Step 2:- Data Reprocessing



Some steps in Data Preprocessing

- Dropping the irrelevant columns is an aspect of data cleaning which is an essential step for data preprocessing.
- The process of scaling data, improves model performance, convergence, and prevents bias from features with larger values.

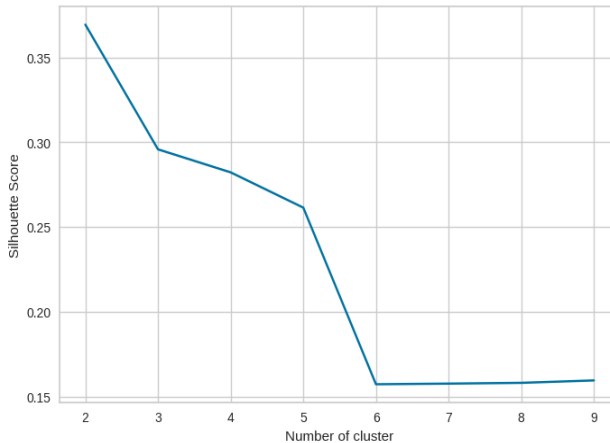
- Data preprocessing/preparation/cleaning is the process of detecting and correcting (or removing) corrupt or inaccurate records from a dataset.
- Feature engineering helps in creating robust models that are less sensitive to noise, outliers, and changes in the dataset. By transforming the features appropriately, models become more resilient to variations, leading to improved performance in real-world scenarios. In case new features such as Age, Family size, Kids, Expense, Engaged in days, number of offers customer accepted etc. were created.
- Consider the percentage of missing values: If the percentage of missing values is small, dropping the rows with missing values may not significantly impact the analysis. However, if the percentage of missing values is high, imputation methods may be necessary. In this scenario, we observe 1% as the percentage of missing value so small that dropping the rows of missing values was a good idea.

Solution Step 3:- Exploratory Data Analytic(EDA)

Applying CPA and t-SNE on data

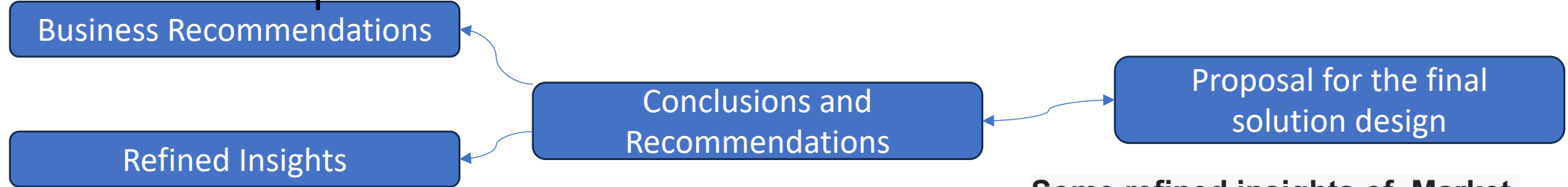
Model building

Cluster profiling



- We applied the CPA and t-SNE clustering methods in this scenario.
 - Combining principal component analysis (PCA) and clustering methods are useful for reducing the dimension of the data into a few continuous variables containing the most important information in the data.
 - Finding the optimal number of clusters is an important part of this algorithm. A commonly used method for finding the optimum K value is **Elbow Method**. In this scenario, the elbow wasn't clearly observed so we used the **Silhouette Method**.
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- Profiling acts as a class descriptor for the clusters and will help you to 'tell a story' so that you can understand this information and use it across your business. For example, group 3 clusters are the highest number of store purchases made by the customer.

Solution Step 4:- Conclusions and Recommendation



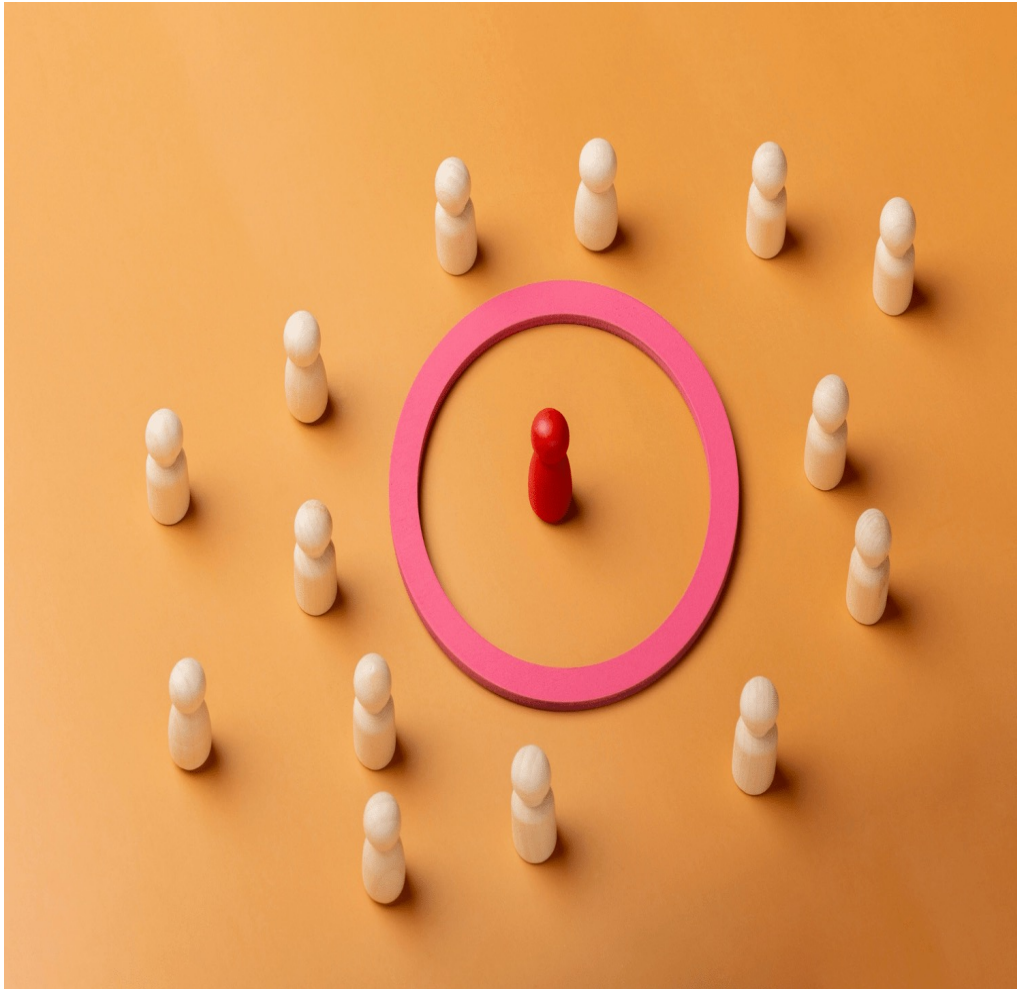
Conclusion. Marketing segmentation is a powerful tool for businesses. It enables an effective understanding and addressing of each customer segment. By integrating these segmentations into your segmentation strategy, you can create more impactful campaigns and strengthen your customer relationships.

Demographic segmentation is one of the most commonly used methods of marketing segmentation. It involves dividing the market into distinct groups based on demographic data such as gender, age, income, or family status as seen in this scenario. For instance, a company that sells beauty products, may have specific items for men and women, or a toy brand might target specific age groups.

Some refined insights of Market Segmentation

- Define a Target Market and Generate Buyer Personas,
- Develop a Marketing Mix Strategy,
- Leverage Data-driven, Personalized Advertising and Marketing,
- Improve Customer Retention and Reduce Churn,
- Create an Informed, Development Process For New Products.

RECOMMENDATIONS FOR IMPLEMENTATION



We recommend to add more values on how to the segment customers such as:

- Review dataset and market analysis,
- Examine your current customer base,
- Choose a customer segmentation model,
- Consider customer segmentation software,
- Collect customer experience data,
- Analyze customer experience data,
- Refine your customer segments.

- Implementing a Better Market Segmentation Plan like Objective Setting. Set segmentation objectives and goals.
 - Identify segmentation variables and develop hypothesis.
 - Identify Customer Segments.
- Develop Segmentation Strategy. Select target segment.
- For example,
 - consider whose customer needs your products,
 - gather data of your customers(Data collection),
 - Look for underserved segments,
 - Research audience behaviors(research design),
 - Develop buyer personas,
 - Consider positioning options,
 - Review your profit potential,
 - Study the competition.

THANK YOU